

Sprint 2 Report (October 5th — November 5th)

YouTube Link of Sprint 2 Video

YouTube video: <https://www.youtube.com/watch?v=iNyf0rqXSvc>

What's New (User Facing)

- 20 Episode Loop Functionality
 - Scripted Policy
 - Scripted Policy Test Courses
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Work Summary (Developer Facing)

For this sprint we started by creating the test courses that would give us the base pass / fail rate for the bot. Then we created the scripted policy for the bot. This policy included no learning and instead relied on action settings to define how the bot behaves in specified scenarios.

We set action parameters that determined bot movement/look speed and the frequency in which these actions were sent to the client. This prevented stutter and allowed the bot to appear more natural. After creating the scripted policy, the bot challenged each test course and the pass/fail data was recorded to a csv named after the respective course.

The final part of the sprint involved optimization that reduced latency and improved reaction responsiveness.

Unfinished Work

The only unfinished component would be the optimization for full player-like movement. This will be refined during Sprint 3 when the learning policy replaces scripted.

Completed Issues / User Stories

- <https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/issues/10>
- <https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/issues/11>
- <https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/issues/12>

- <https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/issues/13>
 - <https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/issues/14>
 - <https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/issues/15>
 - <https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/issues/16>
 - <https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/issues/17>
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Incomplete Issues / User Stories

N/A

Code Files for Review

- [https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint 2/ai_agent_project/ai/src/app/server.py](https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint%202/ai_agent_project/ai/src/app/server.py)
 - [https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint 2/ai_agent_project/ai/src/app/policy_worker.py](https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint%202/ai_agent_project/ai/src/app/policy_worker.py)
 - [https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint 2/ai_agent_project/ai/src/policy/Goal Nav Policy.py](https://github.com/Strider3039/CAPSTONE-Minecraft-Ai-Agent/blob/Sprint%202/ai_agent_project/ai/src/policy/Goal_Nav_Policy.py)
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Retrospective Summary

Went Well

- Scripted Policy Test Course Creation
- Evaluation Loop Creation
- Evaluation Loop Log

Needs Improvement

- Smoothness of Bot Movement
- Project Latency

Next Sprint Changes

- Implement Learning Policy (delete scripted)
- Implement full player functionality (inventory management, craft / build / mine)